Does Idiomatic Phrasing Degrade LLM Task Performance? A Controlled Study Across Instruction-Tuned Decoder Models

Lihu Zur (322216151), Assaf Haik Barouch (206783441)

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1 Introduction

Idiomatic expressions, whose meaning cannot be derived compositionally from their individual words (e.g., *"spill the beans"* or *"par for the course"*), are routinely treated in NLP as a harder case for language models than direct, literal phrasing. The premise is intuitive: idioms are culturally specific and non-compositional, so a model must have learned an expression rather than derived it, and models do degrade on expressions they have not seen [1]. It is rarely tested *in isolation*, though. Any two rewordings of a sentence differ lexically and syntactically, so a drop after "idiomatizing" one could be an artifact of rewording in general.

Central research question: does phrasing a task input idiomatically cause a measurable degradation in a language model's task accuracy, *over and above* the degradation that any comparable paraphrase (a non-idiomatic rewording) already causes?

We rewrite every original example x into two variants sharing a rewriting budget but differing in one property: $x_{\text{paraphrase}}$, reworded **without** idioms, and $x_{\text{idiomatic}}$, reworded **with** them. The label y is given to the rewriter and re-checked afterwards, failures being dropped rather than relabelled (§2.2), and each variant is evaluated under an identical prompt template.

The answer decides where effort should go. Figurative phrasing is a property of the incoming data wherever informal text reaches a classifier rather than being authored for one: review sentiment, ticket triage, content moderation. If idiomaticity has an isolable effect there, that bears on prompt engineering, augmentation and robustness evaluation; if idiom failures are merely generic paraphrase sensitivity, work specific to idioms is misplaced.

Contributions. We contribute (i) the paraphrase-controlled design above, instantiating for a linguistic property the control-condition logic concurrent work establishes in general [2]; (ii) a *sole-cause* diagnostic that, restricted to items a model already answers correctly, identifies which single rewrite is responsible when exactly one of the two flips the answer; (iii) dual significance testing, pairing continuity-corrected McNemar with a non-parametric bootstrap; (iv) breadth, 14 instruction-tuned decoder LLMs from 0.5B to 7B across 10 families on three datasets, for 42 runs with a pooled cross-model test per dataset; and (v) a reproducible four-stage pipeline with LLM-judge validation of every augmented example and per-artifact provenance (§2).

1.1 Related Work

Idiom processing in NLP. Tayyar Madabushi et al. [1] probe pretrained models on the literal-versus-idiomatic distinction, finding they do well given a few labelled examples per expression but degrade substantially zero-shot on unseen ones, and Chakrabarty et al. [3] introduce FLUTE, a 9,000-instance figurative-language benchmark. The LLM era inherited this framing: Phelps et al. [4] find prompted LLMs short of fine-tuned models even at GPT-4 scale, and Mi et al. [5] show LLMs failing to resolve idiomaticity when doing so depends on context. The weakness appears outside classification too: Golan Hashiloni et al. [6] introduce IdioLink, a retrieval benchmark of 2,140 queries over 107 idioms, and find embedding models matching on surface form instead of linking an idiom to a literal paraphrase of the same meaning. All of this measures idiom *comprehension*, not what idiomatic phrasing costs on a task that is not about idioms.

Figurative language inside a task. Stowe et al. [7] pair idiomatic and metaphoric sentences with literal

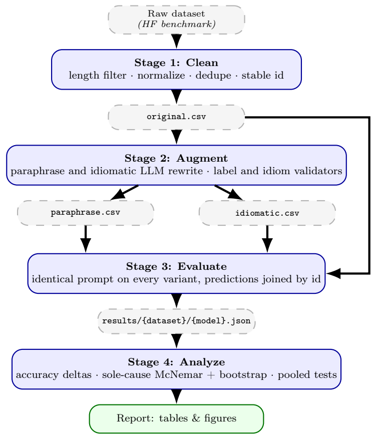

Figure 1: The four-stage pipeline, instantiated once per dataset/model combination. Raw data is cleaned (Stage 1), rewritten into validated paraphrase and idiomatic variants (Stage 2), evaluated per model on every variant under an identical prompt (Stage 3), and aggregated into the accuracy-delta and sole-cause analysis (Stage 4). `original.csv` feeds both Stage 2, as the rewriting source, and Stage 3, as the control variant. The orchestrators carry no dataset- or model-specific branching; each dataset adds its own loader and evaluator module.

counterparts and find MNLI-finetuned RoBERTa recognizes figurative–literal entailment yet fails on matched non-entailing pairs. For metaphor, Sanchez-Bayona and Agerri [8] conclude from NLI and QA experiments that LLM performance is shaped more by lexical overlap and sentence length than by metaphorical content, consonant with ours. Both annotate the figurative property where it already occurs; we introduce it by controlled rewrite of items whose original form is also scored, which is what makes a matched control arm possible.

Why a paraphrase control is necessary. Models are sensitive to edits that leave the correct answer intact: Ribeiro et al. [9] flip predictions by *rewording* inputs under semantics-preserving rules, and Cao et al. [10] find Llama-2-70B-chat's performance varying by 45.5 points between the best and worst of a set of semantically equivalent queries. A bare original-versus-idiomatic contrast therefore measures idiomaticity confounded with rewording sensitivity.

Controlled evaluation design. Both our rewrites are invariance tests in the sense of CheckList [11], the mirror image of contrast sets [12], which perturb minimally to *change* the gold label. What we add is subtraction against a matched control, the move probing classifiers make with control tasks [13]. Control conditions exist in figurative evaluation already, but as answer-space distractors rather than arms whose accuracy is subtracted: MUNCH's deliberately inapt paraphrases detect lexical-similarity shortcutting [14].

Yang et al. [2] argue the general case, reporting that on MedQA a 14.9% flip rate under a demographic edit is statistically indistinguishable from the 14.1% induced by paraphrasing alone. We instantiate that logic for a linguistic intervention, and unlike the demographic case a residual does survive on inference. Studies injecting a property without such a baseline, such as the dialect rewrites of GLUE by Ziems et al. [15], bound its contribution from above rather than estimating it. To our knowledge idiom-inserted rewrites of general-purpose NLU benchmarks have not previously been scored against a matched paraphrase arm over the same items.

2 Methodology

2.1 Pipeline overview

The project is implemented as four independently re-runnable stages (Figure 1) communicating only through versioned files on disk, plus YAML configs setting each stage's run parameters; per-dataset modules hold the loading and answer-parsing logic. Any stage can be re-executed alone, so evaluating a new model does not require repeating cleaning or augmentation, and every artifact records the config hash and model/prompt version that produced it.

Stage 1: Clean. Each raw dataset is loaded from Hugging Face at a pinned revision, an unpinned dataset being a hard error, and reduced to one clean CSV of rows suitable for rewriting. Four filters run in a fixed order, identically configured for all three datasets except where noted:

Table 1: One aligned *sole-cause* triple per dataset, verbatim from `datasets_out/`: the original and paraphrase were answered correctly and the idiomatic variant was not, so the idiom alone broke the answer. Only one field per dataset is rewritten; everything else is byte-identical across rows. **Bold** marks the aligned span each variant puts in the same role.

MNLI – hypothesis: *"No one would give them credit like that."*; gold label *contradiction*. Idiomatic-only-wrong for 12 of 14 models.	
original	Even the Scots are **likely to give them more credit** than that.
paraphrase	Even the residents of Scotland **will probably accord them a higher level of recognition** than that amount.
idiomatic	Even the Scots are **unlikely to sell them short** like that.
SST-2 – gold label *negative*. Idiomatic-only-wrong for 10 of 14 models.	
original	in which the rest of the cast was **outshined** by ll cool j.
paraphrase	in which the rest of the cast was **outperformed** by ll cool j.
idiomatic	in which LL Cool J **stole the show** from the rest of the cast.
MMLU – gold answer *"The nation invests in research and development of new technology."*; the four choices are untouched. Idiomatic-only-wrong for 8 of 14 models.	
original	Which of the following scenarios would **increase** a nation's production possibility frontier (PPF)?
paraphrase	Which of the following situations would **expand** a country's production possibility frontier (PPF)?
idiomatic	Which of the following scenarios would **push out** a nation's production possibility frontier (PPF)?

1. **Normalize.** Unicode NFC, whitespace collapsed to single spaces.
2. **Length band.** The field to be rewritten must hold 10 to 100 whitespace-separated tokens: shorter leaves no room for an idiom, longer costs more to augment and, we assumed without testing it, dilutes a fixed-size rewrite.
3. **Deduplicate.** Rows colliding on the full (normalized text, label, metadata) triple are dropped, keeping the first.
4. **Shuffle and cap.** Survivors are shuffled under a fixed seed, then cut to a 2,000-row cap; MNLI is uncapped. It binds only on SST-2 (25,901 rows cut to 2,000); MMLU's 1,107 already fall under it.

Each retained row then gets a stable identifier aligning all its downstream variants; the label y and dataset metadata (e.g., MMLU's answer choices) carry through unchanged.

2.2 Augmentation into paraphrase and idiomatic variants

Stage 2: Augment. Each cleaned row is rewritten by an external LLM (Google Gemini) into two variants using two fixed prompt templates. One Gemini sub-model is pinned per dataset for cost reasons (`gemini-3.6-flash` for SST-2 and MMLU, `gemini-3.5-flash-lite` for MNLI) and shared by both arms, so the paraphrase control stays internally valid within a dataset; only magnitudes *across* datasets are not cleanly comparable. Templates are selected per dataset so the wording suits the input type, MNLI using its own NLI-specific pair.

The *paraphrase prompt* asks for a rewording in different words that explicitly avoids idiomatic expressions; the *idiomatic prompt* asks for one that naturally incorporates idioms or figures of speech while preserving meaning, register and length. Both are given the task label y, so a rewrite cannot silently invert the correct answer. An external LLM is used rather than one under evaluation, requiring its idiom judgment to be at least as reliable as the systems tested; we did not benchmark that against them.

Every augmented row then passes a validator chain: **label preservation**, an LLM-judge check that the rewrite has not altered the correct label, and **idiom presence/absence**, a per-variant LLM-judge check requiring a genuine idiom in the idiomatic variant and none in the paraphrase. Rows failing validation after bounded retries are dropped from that variant and recorded in a sidecar manifest, which is why the rewritten CSVs are smaller than the original. An on-disk cache keyed by (prompt hash, augmenter model, input id) makes reruns idempotent. Table 1 shows one produced triple per dataset.

2.3 Datasets

Three datasets are evaluated, each scored with the accuracy metric of its original publication. Stage 2 rewrites one field per dataset (SST-2's sentence, MMLU's stem, MNLI's premise); everything else, including MMLU's answer choices and MNLI's hypothesis, stays byte-identical across the three variants.

SST-2 [16] is binary sentiment classification over short movie-review snippets. **MMLU** [17] is multiple-choice knowledge questions spanning 57 subjects. **MNLI** [18] asks whether a premise entails, contradicts, or leaves undetermined a paired hypothesis, over sentence pairs drawn from ten written and spoken genres.

The evaluated split differs by dataset, for one reason: every scored row needs a gold label, so a split whose labels are withheld is unusable however large. MMLU therefore uses its `validation` split, MNLI its two validation splits (below), and SST-2, whose GLUE test labels are withheld, draws from `train` and `validation` combined. SST-2's 2,000-row cap is a compute-budget decision, unrelated to that choice.

The three cover qualitatively different task types under one protocol: short, informal, already-somewhat-figurative text; longer, formal multiple-choice questions; and sentence-pair inference.

They also differ in how the rewritten field sits relative to the decision, which is a property of the design rather than a prediction. In SST-2 the rewritten sentence *is* the object being classified, and sentiment is a property an idiom may carry rather than obscure. In MMLU only the stem is rewritten, while the four answer choices, where the knowledge tested sits, stay untouched. In MNLI the rewritten premise is the span the label is computed from, the hypothesis being fixed. §4 returns to whether this explains what we observe. Separately, chance sits at 50% for SST-2, 33% for MNLI and 25% for MMLU. Because the sole-cause test counts only rows answered correctly on the original (§2.5), the share of rows it can draw on tracks that floor: pooled across models, 89% on SST-2, 74% on MNLI and 53% on MMLU. MMLU therefore works from much the smallest base.

MNLI-specific protocol. MNLI pairs roughly three hypotheses with every premise; emitting all would repeat the premise and violate the independence assumption behind the paired test, so the loader keeps only the first hypothesis per premise. Rows come from the two validation splits combined (matched, 9,815 pairs; mismatched, 9,832), the mismatched half contributing the five genres absent from training. Each row's genre is recorded but not split on. After first-hypothesis selection and cleaning, 5,127 premises survive, of which 4,709 (91.8%) also clear validation in *both* rewritten variants and so form the aligned triples.

2.4 Evaluated models

Rather than a single-model case study, we evaluate a panel of 14 open, instruction-tuned decoder LLMs spanning 0.5B to 7B parameters across 10 model families: Qwen2.5-Instruct (0.5B, 1.5B, 7B), Falcon3-Instruct (3B, 7B), OLMo-2-Instruct (1B, 7B), Gemma-2-2B-it, Mistral-7B-Instruct-v0.3, Phi-4-mini-instruct (3.8B), Granite-3.1-2B-Instruct, H2O-Danube3-4B-Chat, Yi-1.5-6B-Chat and SmolLM2-1.7B-Instruct. Tables and prose below drop the `-Instruct`/`-it`/`-Chat` suffix. All are evaluated zero-shot under an identical dataset-specific prompt template applied to all three variants of a row, so only the input text differs between runs; a device-agnostic runner selects CUDA, Apple-Silicon MPS or CPU and adjusts precision accordingly.

Stage 3: Evaluate. For each (model, dataset) pair the model is run on all three variant CSVs with the identical prompt, and predictions are joined by row id, yielding one matched (original, paraphrase, idiomatic) triple per surviving row. Per-variant accuracy is computed from these joined predictions, scoring as incorrect any generation the dataset's parser cannot map to a label, a choice §3 re-tests on the subset where all three parsed.

2.5 Statistical framework

Stage 4: Analyze. Two families of statistic are computed. The first is per-variant accuracy and the two deltas between variants, reported throughout in percentage points (pp):

$$\Delta_{\text{par}} = \text{acc}_{\text{par}} - \text{acc}_{\text{orig}}, \qquad \Delta_{\text{idi}} = \text{acc}_{\text{idi}} - \text{acc}_{\text{par}}. \tag{1}$$

Δ_{par} measures the cost of rewording and Δ_{idi} the additional cost of idiomaticity on top of it; the research question of §1 is precisely whether $\Delta_{\text{idi}} < 0$. Each is accompanied by a paired-bootstrap confidence interval and a McNemar exact p-value over the corresponding discordant pairs.

The second, and our primary diagnostic, is a *sole-cause* test, which attributes degradation to a single rewrite type rather than to noise. It restricts attention to rows where all three variants exist and the model answered the **original** correctly, so any later error is attributable to the rewrite rather than to a hard item, then counts b = *idiomatic-only-wrong* and c = *paraphrase-only-wrong*. Rows where both rewrites agree carry no information about which is worse and are excluded, as in any paired test. The gap between b and c is assessed two ways:

McNemar's test [19], the standard test for paired nominal comparisons, with the continuity correction of

Table 2: Per-dataset summary over aligned triples. Δ = mean pp across models; *sig.* = models with significant negative/positive Δ_{idi} ($p < 0.05$, McNemar exact). *Parse-clean* repeats both after dropping triples with any parse failure. Pooled sole-cause sums b and c across models, tested by continuity-corrected McNemar and a 10,000-resample bootstrap.

			All triples			Parse-clean		Pooled sole-cause			
Dataset	Models	n	Δ_{par}	Δ_{idi}	sig.	Δ_{idi}	sig.	b/c	χ^2	p_{χ^2}	p_{boot}
SST-2	14	1,548	−1.28	+1.35	0/5	+0.76	0/3	533/723	28.4	<0.0001	<0.0001
MMLU	14	1,085	+0.13	−0.79	1/0	−0.77	1/0	474/378	10.6	0.0011	<0.0001
MNLI	14	4,709	−0.68	−1.28	10/0	−1.26	10/0	2,804/2,246	61.4	<0.0001	<0.0001

Table 3: Per-model Δ_{idi} (pp), ordered by parameter count, two panels. **Bold**: Δ_{idi} itself significant ($p < 0.05$, McNemar exact). †: the separate sole-cause b vs. c gap is significant on *both* McNemar and bootstrap (either direction). The two marks are independent, so a cell may carry either, both, or neither. Negative values support the hypothesis; SST-2 is almost entirely positive.

Model	SST-2	MMLU	MNLI	Model	SST-2	MMLU	MNLI
Qwen2.5-0.5B	+0.45	−1.75	−0.40	Phi-4-mini-3.8B	−0.19	−1.84	**−1.51**†
OLMo-2-1B	+1.29	−0.28	**−1.70**†	H2O-Danube3-4B	**+1.74**	**−2.49**	+0.04
Qwen2.5-1.5B	+1.16	−2.03†	**−1.59**†	Yi-1.5-6B	+0.97	−0.18	**−1.89**†
SmolLM2-1.7B	**+2.20**	−0.74	+0.30	Qwen2.5-7B	**+3.04**†	+0.09	**−1.17**†
Gemma-2-2B	+1.36†	−1.38	**−1.51**†	Falcon3-7B	−0.26	+0.65	**−1.23**†
Granite-3.1-2B	**+1.49**†	+2.12	−0.89	Mistral-7B-v0.3	**+4.13**†	−1.20	**−2.08**
Falcon3-3B	+0.39	−0.37	**−2.25**†	OLMo-2-7B	+1.16	−1.66†	**−2.04**†

Edwards [20]:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \qquad \mathrm{df} = 1, \tag{2}$$

testing the null hypothesis that b and c are drawn from the same underlying 50/50 rate (i.e., among discordant rows, it is a coin flip which rewrite broke it).

Non-parametric bootstrap check. Because that chi-square approximation can be unreliable when $b + c$ is small, every result is cross-checked with a paired bootstrap [21]: the aligned rows are resampled with replacement 10,000 times, $b - c$ is recomputed on each resample, and a 95% percentile interval plus a two-sided empirical p-value are read off. A result counts as robustly significant only when both checks agree, **as they do in all 42 per-model runs**.

The same statistics are additionally pooled across models per dataset (summing b and c, resampling the concatenated rows), with the caveat that this is not a rigorous mixed-effects test and is reported as indicative only (§4.1).

3 Experimental results

The sweep covers 42 (model, dataset) runs, all 14 models on each of the three datasets, and so 126 variant-level inference passes. Every number below is computed over *aligned triples*: rows where the original and both validated rewrites exist and parsed, leaving 1,548 rows for SST-2, 1,085 for MMLU and 4,709 for MNLI. Stage 3 also scores the full original files (2,000/1,107/5,127 rows), which have no counterpart to pair against.

One manipulation, three directions (Table 2). The same idiomatize-a-matched-paraphrase intervention produces a significant *penalty* on MNLI (10/14 models, pooled sole-cause $p < 10^{-4}$), a significant *benefit* on SST-2 (5/14 positive, none negative, pooled sole-cause significant in the opposite direction), and a marginal penalty on MMLU that survives only in aggregate (1/14 individually, $p = 0.0011$ pooled). Idiomaticity is therefore not a general accuracy tax. Any account of it has to explain a change of sign across tasks, not merely a change of magnitude; §4 considers one candidate.

MNLI: a consistent idiom-specific penalty. Ten of 14 models show a significant penalty, from Falcon3-3B at −2.25 pp to Qwen2.5-7B at −1.17 pp (Table 3), and the sole-cause test agrees: $b > c$ for 12 of 14, both tests concurring at $p < 0.05$ for 9 of the same 10 significant models. Only Mistral-7B-v0.3 splits, with a significant accuracy delta but a non-significant sole-cause gap ($b = 225$, $c = 191$). Nine models agreeing on two tests with different evidence bases is the strongest single result in this study. Table 4 works one such row through in full for a significant model on each dataset.

Table 4: Worked disagreement cases: three rows where a model with a *significant* Δ_{idi} predicted differently under the paraphrase and the idiomatic rewrite (✓ correct, × wrong). **Bold** marks the aligned span. Every variant is scored under the same fixed system prompt, so only the rewritten field changes within a block. MNLI and MMLU show the majority direction, the idiom alone breaking a correct answer; SST-2 shows that dataset's reversal.

Variant	User prompt	Prediction
MNLI – Falcon3-3B-Instruct ($\Delta_{\text{idi}} = -2.25$ pp). Every prompt ends `Hypothesis: She now knows what was going on in her mind. / Answer:`. Gold: *entailment*.		
original	`Premise:` It took me really a long time to **sort** what was going on in my mind **out**, she said.	entailment ✓
paraphrase	`Premise:` She stated that it required a significant amount of time for her to **organize her thoughts**.	entailment ✓
idiomatic	`Premise:` It took me quite a while to **wrap my head around** what was going on in my mind, she said.	contradiction ×
SST-2 – Mistral-7B-Instruct-v0.3 ($\Delta_{\text{idi}} = +4.13$ pp). Gold: *positive*.		
original	`Sentence:` is n't embarrassed to **make you reach for the tissues**	positive ✓
paraphrase	`Sentence:` is not ashamed to **make you cry**	negative ×
idiomatic	`Sentence:` isn't shy about **pulling at your heartstrings** and making you reach for the tissues	positive ✓
MMLU – H2O-Danube3-4B-Chat ($\Delta_{\text{idi}} = -2.49$ pp). The four choices are untouched: `A.` a logical mistake. `B.` an ethical mistake. `C.` a factual mistake. `D.` not a mistake. Gold: *D*.		
original	`Question:` **According to Nagel**, the view that moral luck is paradoxical is:	D ✓
paraphrase	`Question:` **Based on Nagel's analysis**, the perspective that moral luck is a paradox is:	D ✓
idiomatic	`Question:` **In Nagel's eyes**, the view that moral luck is paradoxical is:	A ×

SST-2: the idiom recovers what the paraphrase cost. Original accuracy is high and tight (83.8–91.7%); plain paraphrasing already costs 12 of 14 models, significantly for 4, and idiomatizing that same rewording recovers most of the loss: positive for 12 of 14, significant for 5, with Mistral-7B-v0.3 swinging widest (−2.91 pp then +4.13 pp). Part is artifact, quantified below; a residual survives that filtering. One reading is that idioms here carry sentiment rather than obscure it, but nothing we ran isolates that from other explanations.

MMLU: a small effect, cause untested. MMLU spreads models much more widely (29.7–68.7%, against a 25% floor) while both deltas stay small (+0.13/−0.79 pp); only H2O-Danube3-4B reaches individual significance, but the sole-cause direction is consistent enough ($b > c$ for 9 of 14) that pooling turns it significant. One structural feature may cap the size available, on no ablation of ours: Stage 2 rewrites only the stem, leaving the four answer choices untouched, so an idiom can obscure what is asked without reaching what the model must know. Models near the chance floor also generate too few sole-cause events to have power (OLMo-2-1B: $b = 8$, $c = 11$).

The SST-2 reversal is partly a parsing artifact. Unparseable generations score as incorrect, so a variant provoking more format violations is penalised for reasons unrelated to idiom comprehension. Failure rates are low on average (1.0%/0.2%/0.3% for SST-2/MMLU/MNLI) but uneven, reaching 6.7% for Mistral-7B-v0.3 on SST-2, and there the *paraphrase* arm fails more often than the idiomatic arm for 11 of 14 models, inflating Δ_{idi}. Dropping every triple with any parse failure leaves MMLU and MNLI untouched, all 10 significant MNLI models included, but cuts SST-2's mean from +1.35 to +0.76 pp and its significant count from 5 to 3. Qwen2.5-7B is the clearest case: an apparent +3.04 pp advantage becomes −0.21 pp once its 6.4%-vs-2.1% parse-failure gap is removed. The direction survives regardless, no SST-2 model showing a significant idiom *penalty* even filtered.

No capability or scale trend. Δ_{idi} correlates with neither original accuracy (−0.13/+0.28/−0.29) nor parameter count (+0.10/+0.24/−0.30) at any significance ($p > 0.3$ throughout). The Qwen2.5 ladder shows it concretely: MNLI's penalty is non-monotone across its three sizes (−0.40, −1.59, −1.17 pp at 0.5B/1.5B/7B), and all four 7B models sit inside the degraded group. Rewriting is not purely destructive either: pooled across models, 419 SST-2, 235 MMLU and 1,614 MNLI rows were wrong on the original but right on both rewrites.

4 Discussion

The control arm is what makes the data legible. The choice doing this work is Eq. 1: Δ_{idi} is measured against *paraphrase* accuracy, not original accuracy, so subtracting the paraphrase arm removes the rewording cost both rewrites incur and leaves only what idiomaticity adds. Score idiomatic against original instead, as a study without a control arm must, and SST-2 looks like a null (+0.07 pp, 1 of 14 significant) while MNLI's cost is overstated by roughly 53%, its naive −1.96 pp bundling in −0.68 pp of plain rewording. The matched arm splits that false null into two real cancelling effects and cuts MNLI's to its idiom-specific size. Neither correction is visible without it.

One candidate explanation, which this study does not test. The ordering (-1.28 MNLI, -0.79 MMLU, $+1.35$ pp SST-2) lines up with how the rewritten field relates to the decision (§2.3): MNLI's premise is the span the label is computed from, MMLU's stem leaves the knowledge in untouched choices, and SST-2's sentence carries the very property being classified, so an idiom there may supply sentiment rather than obscure it.

That is a hypothesis suggested by the results, not something the design tests. Three datasets cannot separate it from the other ways they differ, and MNLI's augmenter and rewrite depth are not matched to the other two, so the ordering is a ranking rather than a magnitude comparison. Its merit is falsifiability; §4.2 gives the experiment. Without it the data still support something firmer: idiomaticity has an isolable effect whose sign is not constant across tasks. That tempers without contradicting the comprehension weaknesses Tayyar Madabushi et al. [1] and Chakrabarty et al. [3] report, since we measure the residual cost after a model applies whatever partial idiom understanding it has, on a task never asked to reason about idioms.

The magnitudes are real but modest. A 1–2 pp MNLI drop is statistically robust, but it is the same order as plain rewording's own cost rather than an order above it. Idiomaticity is best read as one distinguishable contributor to paraphrase sensitivity, not as a separate failure mode.

Scale should not be assumed to fix it. Across a 14-fold parameter range no correlation with model size exceeds $|\rho| = 0.30$, but that is weaker than it looks: with 14 models, $|\rho|$ must reach about 0.53 before the test can call anything significant, so a moderate trend could be present and go undetected. Nothing in this panel supports waiting for a larger checkpoint to solve the problem, and settling the question needs many more models than 14.

4.1 Threats to validity

The parse-failure filter is blunt. It drops 2.2% of SST-2 rows on average but 12.4% for Mistral-7B-v0.3 there, and 8.2% for OLMo-2-1B on MNLI. Dropping rows non-randomly can itself bias a comparison, so the filtered and unfiltered readings bracket the truth rather than either being correct.

The pooled tests overstate their own confidence. Pooling sums b and c across models and resamples the concatenated rows, treating every (model, row) pair as independent. They are not: each row recurs once per model, so a row that is idiom-hard for many models is counted many times over. Pooled intervals therefore come out narrower, and pooled p-values smaller, than the evidence warrants. Correcting it needs a mixed-effects model with a random effect per row, which we did not run, so pooled figures are indicative only. That falls hardest on MMLU, whose result rests on pooling alone.

No multiple-comparisons correction is applied. Fourteen tests at $\alpha = 0.05$ expect 0.7 hits under a true null, so MMLU's single significant model carries little weight ($P(X \geq 1) = 0.51$). MNLI's 10 and SST-2's five are clear of chance (8×10^{-11} and 4×10^{-4} respectively).

Scope limits. Every model was evaluated zero-shot, greedily, under a single prompt template per dataset. Few-shot prompting, sampled decoding and prompt-format variation are therefore untested and could interact with idiomaticity. The panel is also decoder-only.

4.2 Future work

Shore up the existing result. The most valuable next step sharpens the MNLI finding rather than broadening it: a direct check that the rewrites preserve meaning, ruling out semantic drift. A mixed-effects re-analysis would then put the pooled claims on the footing §4.1 says they lack.

Test the explanation rather than assume it. Two analyses reuse the rewritten text already collected, but both need a step we skipped: Stage 2 records no inventory of which idiom it inserted where, so each row's idiom must first be identified. The first then grades that idiom's opacity against published psycholinguistic norms and correlates it with Δ_{idi}; if the cost comes from non-compositionality, more opaque idioms should cost more, whereas today every idiom counts the same. The second does the equivalent for frequency from corpus counts, where rarer idioms costing more would confirm on our task the familiarity account Tayyar Madabushi et al. [1] establish for comprehension, which we can currently only invoke by analogy. Scoring either through the augmenter or model perplexity instead needs fresh inference.

A third, more expensive experiment tests the hypothesis of §4 directly, varying idiom position *within* one dataset rather than across three: rewriting MNLI's hypothesis as well as its premise, or MMLU's choices as well as its stem, makes position a manipulated variable with dataset, augmenter and models fixed. It needs new augmented data and a fresh sweep.

Try to reduce the penalty, not only measure it. Everything above is diagnostic. Few-shot prompting is the cheap intervention: in-context examples may recover the accuracy idioms cost, as labelled examples do for the fine-tuned encoders of Tayyar Madabushi et al. [1]. Demonstrations also constrain output format, so they should cut the unparseable-generation rate §3 shows contaminates the SST-2 comparison, separating format from comprehension there. Few-shot prompting the *augmenter* should likewise cut the validator failures that currently cost us rows. Fine-tuning is the substantial one: briefly tuning each model on labelled idiomatic rewrites would show whether the penalty is a fixable gap in exposure or something a model this size cannot learn away.

5 Summary

We measured the accuracy cost of idiomatic phrasing for 14 instruction-tuned decoder LLMs from 0.5B to 7B parameters on SST-2, MMLU and MNLI, pairing every idiomatic rewrite with a matched non-idiomatic paraphrase of the same row so that the cost of rewording is subtracted rather than absorbed into the effect. That control is what makes the result readable. Without it, SST-2 looks like a null and MNLI's idiom-specific penalty is overstated by roughly 53%. With it, the penalty concentrates on MNLI: -1.28 pp there, significant for 10 of the 14 models, against -0.79 pp on MMLU with a single significant model and no penalty at all on binary sentiment. These are means across the 14 models rather than pooled figures; the pooled sole-cause tests are reported separately in Table 2. Idiomaticity is therefore a distinguishable contributor to paraphrase sensitivity rather than a separate failure mode. No scale trend is detectable across a 14-fold parameter range, though a panel of 14 models could not have detected a moderate one.

6 Code

Code repository: `https://github.com/LihuZur/Idiom_degradation`

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